Air Pollution Control Division
Field Inspection Report

COLORADO Department of Public Health & Environment

APCD RECEIVED 1/25/2019 DP

COUNTY NUMBER: **013** SOURCE NUMBER: **0003**

DATE OF INSPECTION: **December 12, 2018** DATE REPORT SUBMITTED: **January 25, 2019**

COUNTY: **Boulder** INSPECTOR: **Grant McKercher**

COMPANY: **CEMEX Construction Materials South, LLC**

SITE LOCATION: **5134 Ute Highway, Lyons, CO 80540**

MAILING ADDRESS: **Same**

CONTACT PERSON: **Scott Harcus (Environmental Manager)** TIME: **10:30 AM**

TELEPHONE NO.: **303-823-2124** EMAIL: **scotta.harcus@cemex.com**

PERMIT NO.: **95OPBO082**

SOURCE CLASS: Major ☒ SM-80 ☐ Syn Minor ☐ Minor ☐

INSPECTION TYPE: Full Compliance Evaluation ☐ Onsite Evaluation ☒
Partial Compliance Evaluation ☒ Offsite Evaluation ☐

Additional Inspection Records in File? Yes ☒ No ☐

HOURS: Travel & Prep: **3.0** Inspection: **1.0** Report: **2.0** Total: **6.0**

COMPLIANCE STATUS: IN COMPLIANCE ☐ OUT OF COMPLIANCE ☒

INTRODUCTION

On December 12, 2018, significant fugitive dust emissions were observed associated with clinker cooling and transfer activities at the CEMEX Construction Materials South, LLC's cement production facility located at 5134 Ute Highway, Lyons, Boulder County, Colorado ("CEMEX"). On December 12, 2018, Mr. Grant McKercher, Inspector with the Colorado Department of Public Health and Environment – Air Pollution Control Division ("Division") was on site to provide test oversight for a VOC emissions compliance test at the main stack (addressed in a separate report, see 0130003-STK-2019). During the test oversight and at about 11:30 AM, Mr. McKercher observed significant dust emissions from clinker transport elevators and an external clinker drop hood at the main clinker production building. Mr. McKercher met with Mr. Scott Harcus, CEMEX Environmental Manager, for a discussion and investigation regarding the cause of the emissions.

Mr. Harcus and maintenance personnel explained that the emissions were associated with a baghouse used to control emissions associated with the clinker cooler area. Mr. McKercher observed the Magnehelic gauge of a clinker cooler area baghouse, which displayed a differential pressure of about zero. The source explained that the baghouse likely was not pulling the required draft to control the clinker dust emissions associated with the elevators, clinker drop hood, and nearby transfer points. The source explained that in days prior to the event, a work order had been placed for replacement motor parts at a different baghouse in the A-frame building. When asked about addressing the observed emissions, Mr. Harcus suggested using the A-frame baghouse for backup control, but until the component arrived and the other baghouse was repaired, the source would not achieve emissions control with either baghouse. The inspector did not observe any CEMEX personnel conducting visible emissions observations or any logs of the event. Once Mr. Harcus observed the emissions, no corrective actions were immediately taken. Mr. Harcus explained that plant operations would likely not be stopped or changed in order to address the uncontrolled emissions. The inspector requested baghouse maintenance records during the event, but as of the date of this report, no baghouse maintenance logs have been provided. CEMEX clinker processes continued to operate throughout the duration of the inspector's visit through the inspector's departure near 1:00 PM. Neither Mr. McKercher nor the source performed an EPA Method 9 opacity observation while the inspector was on site.

CEMEX called the event into the Division malfunction hotline on 12/12/2018. After guidance was provided, the source also provided a follow-up report of the event via email on 12/17/2018 (see attached). The report lists that the baghouse was identified as Baghouse Duct Collector #525-5 and that a damper was adjusted on 12/13/2018 to resolve the issue.

POINT AIRS ID/PERMIT NUMBERS

CEMEX operates under Operating Permit No. 95OPBO082. The following table includes the emissions points that are relevant to this report, as listed in the permit.

Process (Permit Section)	Plant ID	AIRS ID	Description	Pollution Control Device	Construction Permit
Clinker Cooling and Transfer to Storage for Finish Mill (Section II.10)	P008	008	S017 – Clinker Drag Chains (1 baghouse)	Baghouse (5 total)	12BO444-2
			S018 - Clinker Cooler (2 baghouses, 1 stack)		
			S023 – 529-25 Drag Conveyor (1 baghouse)		
			S024B – Outside Clinker Drop Hood (1 baghouse, vented to S018 stack through 525-8/9)		
Clinker and Gypsum/Additive Silos and Weigh Feeders (Storage and Transfer to Finish Mill) (Section II.11)	P009	009	S021 – Top of A Frame (Belt 529-30 to 529-63)[1]	Baghouse (14 total)	98BO0259
			S026, S027, S029, S030, S031 – Weigh Feeders 1, 2, 4, 5 and 6[1]		
			S024 - #2 Clinker Silo		
			S032 – Bottom of A Frame Transfer[1]		
			S033 Gypsum/Limestone from 529-31 belt to Silos		
			S035 – Discharge of 629-3 Belt		
			S039 to S041 –Finish Mill Weigh Feeders[2]		
			S038 - Surge Bin[2]		
			[1] stacks vent inside A-Frame		
			[2] stacks vent inside mill building.		
Sheltered (A-Frame) Clinker Storage and Reclaim (Section II.11)	P010	010	S034 - #6 Reclaim Feeder and A-Frame Building	Baghouse	98BO0259
			S051 – Top of A Frame – Transfer from 529-29 belt to 529-30 belt		
Outdoor Clinker Piles and Handling (Section II.11)	P015	015	Outdoor Hot Clinker Pile	PM Emission Control Plan	98BO0259

Cement Finish Mill and Auxiliaries (Section II.11)	P011	011	S036 - Finish Mill	Baghouse (2 total)	98BO0259
			S037 – Finish Mill Auxiliary Dust Collector		
			Grinding and Limestone Handling		
	P012	031	S065 – Finish Mill Separator	Baghouses (2 total)	98BO0259
			S069 - Clinker Dust to Finish Mill (SEP project) – vents inside mill room	Baghouse	

SOURCE COMPLIANCE HISTORY

A full compliance history is listed in the 2018 full compliance evaluation report.

The 2018 inspection, conducted on 9/11/2018, found violations of Operating Permit No. 95OPBO082, Section II, Conditions 19 and 20.3 for failing to maintain the P010 area (BH 525-17) baghouse differential pressure within the acceptable range.

NSPS/NESHAP/MACT APPLICABILITY

A full list of applicability is listed in the 2018 full compliance evaluation report.

No NSPS/NESHAP/MACT rules apply to the issue addressed in this report.

REPORTS

Reporting is not applicable for this report.

MALFUNCTION REPORT REVIEW

The event was not a malfunction. No malfunctions reports were submitted to the Division.

COMPLIANCE ASSISTANCE/SOURCE ACTIONS

No compliance assistance was provided.

PERMIT CONDITIONS AND COMPLIANCE STATUS

Operating Permit No. 95OPBO082:

The format of the most recent issuance is followed in this section. Only the conditions that apply are listed. Each condition is from Section II of the permit. Text marked in **bold font** indicates inspector comments for each condition.

19. Baghouse Operation and Maintenance

    Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. Routine maintenance and operational procedures shall be in written format. A copy of the operating and maintenance procedures, schedules for maintenance and/or inspection activities and records related to the operation and maintenance of the baghouses and good engineering practices, such as records of routine maintenance and/or inspections shall be maintained and made available to the Division upon request.

Observations on 12/12/2018 indicated that a baghouse associated with clinker cooling and transfer activities had a differential pressure (DP) that was out of the acceptable range. Heavy fugitive dust emissions were associated with equipment's failure to capture emissions. This event was an indicator of a failure to

perform maintenance of and operational procedures in accordance with manufacturer's specifications and/or good engineering practices.

The source is <u>out of compliance</u> with this condition.

20. Colorado Regulation No. 1 Opacity Requirements

20.1 Except as provided in Condition 20.2, below, no owner or operator of a source shall allow or cause the emission into the atmosphere of any air pollutant which is in excess of 20% opacity. This standard is based on 24 consecutive opacity readings taken at 15-second intervals for six minutes. The approved reference test method for visible emissions measurement is EPA Method 9 (40 CFR Part 60, Appendix A (July, 1992)) in all subsections of Section II.A of Regulation No. 1. (Colorado Regulation No. 1, II.A.1).

20.2 No owner or operator of a source shall allow or cause to be emitted into the atmosphere any air pollutant resulting from the building of a new fire, cleaning of fire boxes, soot blowing, start-up, any process modification, or adjustment or occasional cleaning of control equipment, which is in excess of 30% opacity for a period or periods aggregating more than six minutes in any sixty consecutive minutes (Colorado Regulation No. 1, Section II.A.4).

Compliance with these opacity limits shall be monitored as follows:

20.3 Baghouses shall be operated and maintained in accordance with the requirements in Condition 19.

Observations on 12/12/2018 indicated that a baghouse associated with clinker cooling and transfer activities had a differential pressure (DP) that was out of the acceptable range. This is a violation of Condition 19 and is thus also a violation of this condition.

The source is <u>out of compliance</u> with this condition.

CONCLUSION

This partial compliance evaluation, based on observations made on 12/12/2018, information provided by the source, and Division records, finds that CEMEX (AIRS ID 013-0003) is out of compliance with the terms and conditions of Permit Number 95OPBO082 due to the following violations. Recommendations related to enforcement action are in **bold font**.

A. Pursuant to Operating Permit Number No. 95OPBO082, Condition 19, CEMEX is required to conduct routine maintenance and operational procedures performed on the baghouses in accordance with manufacturer's specifications and/or good engineering practices. On 12/12/2018, heavy fugitive dust emissions were observed associated with a baghouse's failure to capture emissions, indicating a failure to perform maintenance of and operational procedures in accordance with manufacturer's specifications and/or good engineering practices, violating Operating Permit Number No. 95OPBO082, Condition 19.

B. Pursuant to Operating Permit Number No. 95OPBO082, Condition 20.3, the baghouses are required to be operated and maintained in accordance with the requirements in Condition 19. On 12/12/2018, heavy fugitive dust emissions were observed associated with a baghouse's failure to capture emissions, indicating a failure to perform maintenance of and operational procedures in accordance with manufacturer's specifications and/or good engineering practices, violating Operating Permit Number No. 95OPBO082, Condition 20.3.

CEMEX was out of compliance on 12/12/2018.

Formal enforcement is recommended to address these violations.